

Three-Mode Hamiltonian Shears in $\mathbb{A}^6$: Exact Degree Growth and Cubic Perron Subfamilies

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Abstract

We prove an exact degree formula for a fixed family of canonical polynomial automorphisms of six-dimensional affine space. Let K be algebraically closed of characteristic zero, let $g \geq 8$, and compose the two triangular Hamiltonian shears generated by

$$V(q) = q_1^2 q_2^2 q_3^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 p_3^2 + p_3^g.$$

The mixed supports force a genuinely coupled two-phase degree calculation. We show that the first shear selects the degree-transfer matrix A_g , the second selects B_g , and the complete step is governed by $C_g = B_g A_g$, not by either half-step matrix alone. A strict cone in the ratio variables $x = u_2/u_1$, $y = u_3/u_1$ is invariant under C_g ; strict old-term gaps and positive coefficients in characteristic zero prevent cancellation at every iterate. The third q -coordinate sees the Perron class, so

$$\deg(F_g^n) = e_3^T C_g^n \mathbf{1}, \quad \lambda_1(F_g) = \rho(C_g).$$

The characteristic polynomial of C_g is cubic, and a mod-five residue audit yields infinite cubic Perron subfamilies for $g \equiv 2, 3, 4 \pmod{5}$. The theorem is exact, proof-first, and bounded to this displayed family; no genericity, priority, or classification claim is made.

1 Introduction

Polynomial automorphisms can hide their degree-growth mechanisms inside a composition word. In a product-like setting, the degree of a composition may reflect only the visible degrees of the factors. In a genuinely coupled setting, however, the selected monomials in one step can alter the winner in the next step, and the complete degree recurrence may only appear after two phases. This paper isolates one such coupled but still fully explicit family.

We work over an algebraically closed field K of characteristic zero and study the map $F_g = T \circ S$ on $\mathbb{A}^6$, where $q = (q_1, q_2, q_3)$ and $p = (p_1, p_2, p_3)$. The displayed potentials are

$$V(q) = q_1^2 q_2^2 q_3^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 p_3^2 + p_3^g.$$

They are asymmetric in exactly one pure-power row each. The asymmetry is deliberate: the first shear should push weight into the p -coordinates, and the second shear should pull that weight back into the q -coordinates.

The exact theorem is the following.

Theorem 1.1 (Main theorem). *Let K be algebraically closed of characteristic zero and let $g \geq 8$ be an integer. Define*

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p), \quad F_g = T \circ S.$$

Then:

1. S and T are polynomial automorphisms with subtraction inverses, and F_g is symplectic for the standard form on $\mathbb{A}^6$.
2. The selected phase matrices are

$$A_g = \begin{pmatrix} g-1 & 0 & 0 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 0 & 0 & g-1 \end{pmatrix},$$

and the complete-step matrix is $C_g = B_g A_g$.

3. On the strict cone

$$\mathcal{K}_g = \{u \in \mathbb{R}_{>0}^3 : x = u_2/u_1 \geq 1, y = u_3/u_1 \geq 1, x + y < (g-3)/2\},$$

there are exactly two selector gaps,

$$M_S = (g-2) - 2x - 2y, \quad M_T = (2g-6)x + (g-6)y - 6,$$

and no third face occurs.

4. The cone is C_g -invariant, the phase/carry induction gives $v_{n+1} = A_g u_n$ and $u_{n+1} = C_g u_n$, and positive coefficients in characteristic zero prevent cancellation of the selected support.
5. The third q -coordinate is visible and maximal, so

$$\deg(F_g^n) = e_3^\top C_g^n \mathbf{1} \quad (n \geq 0).$$

Consequently $\lambda_1(F_g) = \rho(C_g)$, and $\rho(C_g) < (g-1)^2$.

6. The characteristic polynomial is

$$\chi_{C_g}(t) = t^3 - (2g+11)t^2 + (g^2 - 16g + 19)t - 9(g-1)^2,$$

and for $g \equiv 2, 3, 4 \pmod{5}$ the mod-five reduction is irreducible, yielding infinite cubic Perron subfamilies.

7. The case $g = 7$ is a strict-cone boundary with no selector tie. No genericity, entropy, universal non-conjugacy, priority, or CAS-based proof certificate is claimed.

The theorem is intentionally linear. It does not branch into an alternative support analysis, a half-step degree observable, or a numerical side computation. The proof first certifies that the two displayed shears are canonical polynomial automorphisms, then freezes the complete support ledger, then proves that exactly two selector margins govern every iterate, and only after those steps turns to the exact degree row and the arithmetic corollary. This order is not stylistic. If the support rows were allowed to drift before the selector analysis, or if the degree observable were guessed before the visibility proof, the statement would no longer be auditable from the displayed family alone.

What makes the family genuinely coupled is that neither phase carries the final degree observable by itself. The first shear creates a p -degree vector whose dominant row is only an intermediate support signal, and the second shear then reads that signal through a different support ledger before returning to the q -coordinates. If one stopped after the first phase, the matrix A_g would describe

Claim	Locked statement	Main-body evidence	Scope qualifier
L1	S, T are polynomial automorphisms and preserve the standard symplectic form	§3.1, Lemma 3.1	formal identity for the displayed family only
L2	the six support rows determine A_g and B_g	§3.2, Lemma 3.2	no omitted monomials or extra rows
L3	exactly two selector gaps occur, namely M_S and M_T	§4.1, Lemmas 4.2–4.3	cone only, $g \geq 8$ only
L4	$C_g = B_g A_g$ preserves the cone through $X' - 1$, $Y' - 1$, and H_2	§4.2, Proposition 4.5	symbolic inequalities, no sampling
L5	the phase/carry induction gives $v_{n+1} = A_g u_n$, $u_{n+1} = C_g u_n$ without cancellation	§5.1–§5.3, Theorem 5.8	characteristic zero only
L6	q_3 is visible and $\deg(F_g^n) = e_3^T C_g^n \mathbf{1}$	§6.1, Proposition 6.2	visible coordinate fixed as q_3
L7	Perron–Frobenius gives $\lambda_1(F_g) = \rho(C_g)$ and $\rho(C_g) < (g-1)^2$	Corollary 6.3	positive primitive matrix only
L8	the cubic characteristic polynomial and the mod-five test produce the Perron subfamilies	§7, Lemma 7.1, Proposition 7.2	arithmetic corollary only
L9	$g = 7$ is a boundary anti-example and the novelty wording is bounded	Remark 4.6, §8	local collision screen only

Table 1: Claims-evidence matrix for the fixed proof ledger. The order L1 $\rightarrow$ L9 is the proof order of the paper.

only a transient support transfer. If one jumped directly to a guessed scalar recurrence, the proof would lose the mechanism that certifies which support row actually survives the iteration. The chain

$$u_n \xrightarrow{S} v_{n+1} \xrightarrow{T} u_{n+1}$$

is therefore not a stylistic decomposition; it is the minimal bookkeeping that keeps the support selection, the old-term dominance, and the final degree observable logically separate.

This also explains why the full-step matrix $C_g = B_g A_g$ enters only after the selector proof. The entries of A_g and B_g are read row by row from the displayed supports, whereas the cone invariance, the visibility argument, and the Perron root belong to the composed matrix C_g . The proof would be circular if it invoked Perron positivity before establishing that the same two selector faces are chosen at every iterate. The manuscript therefore first proves that the selected support is stable and only then uses the positivity of C_g to turn that stable support into an exact recurrence.

Table 1 is part of the argument, not a decorative summary. It records where each locked assertion is proved and just as importantly where it is *not* proved. The degree formula does not appear before the visibility argument, the cubic Perron corollary does not appear before the characteristic polynomial audit, and the $g = 7$ discussion is postponed until the open-cone proof has finished. The draft is therefore organized as a proof ledger rather than as a broad survey of symplectic polynomial automorphisms.

The closest literature is only context. Section 2 records two bounded citations: one for affine-triangular terminology and one for the neighboring four-dimensional degree-growth context. Neither citation is used to prove the theorem above, and no broader literature claim is needed for the argument.

The paper contributes four things, all internal to the fixed family:

1. an explicit six-dimensional canonical gradient shear pair with a complete support ledger and symplectic block check;
2. an exact two-face selector proof, including the corrected M_T and the $g = 8, \dots, 11$ versus $g \geq 12$ cone split;
3. a phase/carry induction that turns selected support into the exact degree row; and
4. a cubic characteristic polynomial whose mod-five reduction yields infinite degree-three Perron subfamilies.

The proof itself moves through four phases. First, Section 3 freezes the maps, their inverses, the block Jacobians, and the support rows. Second, Section 4 proves that only two selector inequalities matter and that the normalized cone is invariant under the full-step matrix. Third, Sections 5 and 6 turn those strict inequalities into a phase-labeled recurrence and then into the exact degree row. Fourth, Section 7 extracts the cubic characteristic polynomial and checks the mod-five residue classes one by one. No theorem critical step is moved to an appendix.

This exactness is the central point of the paper. The theorem is not an asymptotic growth statement, because the recurrence is not estimated from above or below in the large- n limit. Instead, every iterate is described by the same matrix recurrence once the cone is entered. Likewise, the Perron root corollary is not a generic spectral bound; it is the spectral radius of one explicit 3×3 matrix with a fully written characteristic polynomial. That is why the theorem is proof-first rather than heuristic: the two selected faces and the one visible coordinate are enough to determine all later iterates.

Two exclusions should be explicit before the paper proceeds. The first is a class exclusion: the theorem concerns one exact Hamiltonian gradient-shear family, not the whole affine-triangular world. The second is a proof-mode exclusion: nothing below appeals to CAS output, numerical iteration, or a finite search certificate. Every formula used in the proof is written in the main body and can be checked directly from the displayed supports.

2 Bounded related work and scope boundary

Affine-triangular terminology. The larger affine-triangular comparison class is a convenient language for degree-growth problems, but it is not the class studied here. Our maps are canonical gradient shears with symmetric Hessian blocks, and the distinction matters because the symplectic proof is tied to that exact structure (Blanc and van Santen, 2019). More concretely, the present argument needs the Jacobian blocks to be lower- and upper-unitriangular with Hessian entries in the off-diagonal blocks. That feature is automatic for gradient shears and is what makes the six-by-six symplectic identity a formal calculation. We cite this source only as a terminology boundary; the theorem does not transfer from it, and no result from that paper is imported as a substitute for the support analysis below.

The distinction also matters at the level of proof architecture. In a general affine-triangular setting, one often proves degree growth by tracking a family of possible leading terms and then showing that one family eventually dominates the rest. Here the argument is sharper: the support family is fixed in advance, the cone is two-dimensional after normalization, and the only degree choices are the two selector gaps M_S and M_T . That is why the paper does not need the broader machinery usually associated with affine-triangular dynamics. The proof is smaller in scope but more explicit in structure.

Four-dimensional neighboring context. The recent dimension-four affine-triangular algebraic-degree context does not address this specific degree problem (Shao and Sun, 2025). It is useful here only as a dimension boundary for terminology: in the present paper the degree transfer is governed by a 3×3 matrix, the cone has a two-parameter normalization (x, y) , and the characteristic polynomial is genuinely cubic. Those three features separate the current family from that four-dimensional context. The citation is therefore contextual rather than foundational; it supplies no symplectic, selector, recurrence, or proof input for the theorem below.

There is also a conceptual difference in what the polynomial data are doing. The neighboring four-dimensional context studies a different algebraic-degree pattern in a different ambient space. Here the three-mode family is chosen so that a single pure-power row is balanced against a mixed sextic row in each phase. That balance is what creates the exact A_g , B_g , and C_g chain, and it is what makes the cubic Perron corollary possible after the mod-five audit. The paper is therefore not a dimension lift of the neighboring result; it is a different exact family with a different proof spine.

Bounded novelty wording. The novelty wording is intentionally restrained. The present manuscript makes no global priority claim and no exhaustive classification claim. What matters for the proof is the exact support pattern written in the displayed family: three q -modes, three p -modes, one mixed sextic monomial in each potential, and one pure-power row in opposite phases. If a future comparison refined the surrounding novelty language, the theorem below would remain the same because it depends only on the explicit gradients, matrices, cone inequalities, and residue checks written in the main body.

This bounded wording is not evasive; it is part of the proof discipline. It prevents several false enlargements at once. The present paper is not a residue-theoretic paper, not a moduli or ramification paper, not a torus-escape paper, and not a generic dimension-lift theorem. It is one exact six-variable Hamiltonian family with one exact recurrence.

The same restraint governs the shape of the related-work section itself. A context citation is useful here only if it marks a boundary of the claim: what comparison class is being excluded, or what neighboring lower-dimensional phenomenon serves merely as orientation. Once those two tasks are done, additional citation accumulation would not prove a single selector inequality, would not change the support ledger, and would not strengthen the Perron corollary. In a proof-first manuscript, the relevant question is therefore not how broad a survey can be attached to the theorem, but which exact enlargements of the theorem must be ruled out before the symbolic argument begins.

The hard anti-claims can therefore be stated before the proof begins:

- no genericity statement for arbitrary Hamiltonian shears, arbitrary dimensions, arbitrary support words, or positive characteristic;
- no entropy, periodic-point, trace, torus, conjugacy, or classification theorem;
- no absolute priority, firstness, or optimality claim; and
- no empirical, numerical, or CAS-produced certificate.

With those boundaries fixed, the remainder of the paper can treat the displayed family as a closed proof object. Every later section is about verifying the degree mechanism of that object and not about widening the scope of the claim.

The scope boundary also explains why the mod-five argument appears in Section 7 but not in the related-work discussion. The mod-five table is not being used as literature evidence; it is a proof

Nearby paradigm	Why it looks close	Why it is not the present theorem
general affine-triangular degree growth	triangular coordinates and degree transfer language	no symplectic block identity or fixed three-mode Hamiltonian support ledger is forced
lower-dimensional coupled shears	same proof ambition: exact degree growth from an explicit family	the present family has three modes, a two-variable cone, and a genuinely cubic characteristic polynomial
residue or periodic exactness problems	arithmetic language and sharp boundary statements can look similar	the present argument proves degree growth, not periodic classification or residue rigidity
torus or conjugacy questions	dynamical systems vocabulary often invites these enlargements	no torus, entropy, conjugacy, or classification claim is made anywhere in the paper
computer-assisted dominance checks	selector comparisons can tempt finite search or CAS confirmation	every selector, cone, and residue check here is written symbolically in the main body

Table 2: Generic comparison ledger. The table records several natural ways to over-read the theorem and replaces each with the narrower statement actually proved.

Forbidden enlargement	What the paper actually claims
“all canonical shears in any dimension”	one fixed family $F_g = T \circ S$ in $\mathbb{A}^6$ with the displayed potentials
“a general entropy or periodic theorem”	one exact degree recurrence and one Perron-root corollary
“a priority or classification result”	a bounded non-exhaustive distinction from nearby constructions, not a global priority claim
“a computer-assisted certificate”	a symbolic proof from the written support rows, cone inequalities, and mod-five audit

Table 3: Anti-claim ledger. Each forbidden reading is replaced by the narrower statement actually proved in the paper.

device inside the main theorem. Likewise, the bounded novelty wording is not a priority claim. It is a contract on phrasing, ensuring that the manuscript stays within the exact support pattern for which the proof was written.

That scope discipline has a practical consequence for the rest of the paper. Once the displayed family is treated as a closed proof object, every later check is local and auditable: the gradients are written out, the support rows are finite, the cone inequalities are explicit, and the arithmetic corollary is a finite residue table. The manuscript is therefore not asking the reader to trust a broad hidden framework. It is asking the reader to verify a small number of exact identities in a fixed order and to resist enlarging the claim beyond what those identities support.

3 Exact family, gradients, inverses, symplectic blocks, and support ledger

3.1 The maps and their inverses

We write the phase variables as $(q, p) \in K^3 \times K^3$. The two canonical shears are

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p).$$

The family is deliberately asymmetric. The pure power q_1^g appears only in the first potential, while p_3^g appears only in the second. This placement means that the potentially dominant pure row of the first shear is in the q_1 -slot, whereas the potentially dominant pure row of the second shear is in the p_3 -slot. The degree mechanism is therefore not visible inside a single triangular map; it only appears after the support created by the first shear is fed into the second. Their explicit gradients are

$$\nabla V = (2q_1 q_2^2 q_3^2 + g q_1^{g-1}, 2q_1^2 q_2 q_3^2, 2q_1^2 q_2^2 q_3), \quad (3.1)$$

$$\nabla W = (2p_1 p_2^2 p_3^2, 2p_1^2 p_2 p_3^2, 2p_1^2 p_2^2 p_3 + g p_3^{g-1}). \quad (3.2)$$

The corresponding Hessian matrices, which will be used only through their symmetry, are

$$H_V = \begin{pmatrix} 2q_2^2 q_3^2 + g(g-1)q_1^{g-2} & 4q_1 q_2 q_3^2 & 4q_1 q_2^2 q_3 \\ 4q_1 q_2 q_3^2 & 2q_1^2 q_3^2 & 4q_1^2 q_2 q_3 \\ 4q_1 q_2^2 q_3 & 4q_1^2 q_2 q_3 & 2q_1^2 q_2^2 \end{pmatrix},$$

$$H_W = \begin{pmatrix} 2P_2^2 P_3^2 & 4P_1 P_2 P_3^2 & 4P_1 P_2^2 P_3 \\ 4P_1 P_2 P_3^2 & 2P_1^2 P_3^2 & 4P_1^2 P_2 P_3 \\ 4P_1 P_2^2 P_3 & 4P_1^2 P_2 P_3 & 2P_1^2 P_2^2 + g(g-1)P_3^{g-2} \end{pmatrix},$$

where $P = (P_1, P_2, P_3) = p + \nabla V(q)$ is the intermediate p -vector. Displaying these Hessians makes the block computation completely transparent: the only reason the symplectic identity works is that the off-diagonal blocks come from symmetric second-derivative matrices. The displayed formulas also explain why the later degree proof has exactly two competitive rows and not more. In H_V , every off-diagonal entry comes from the mixed sextic $q_1^2 q_2^2 q_3^2$, while only the $(1, 1)$ -entry carries the additional pure-power contribution $g(g-1)q_1^{g-2}$. In H_W , the same asymmetry is mirrored in the opposite corner: the mixed sextic fills the matrix, and only the $(3, 3)$ -entry receives the extra term $g(g-1)P_3^{g-2}$. Thus the first-derivative competition between one mixed row and one pure row is already encoded at the second-derivative level. The block symplectic calculation and the later selector calculation are therefore different views of the same structural asymmetry, not unrelated steps that happen to use the same polynomials. The inverses are the subtraction inverses

$$S^{-1}(q, p) = (q, p - \nabla V(q)), \quad T^{-1}(q, p) = (q - \nabla W(p), p).$$

No denominator, birational correction, or analytic continuation appears. Thus every later degree calculation takes place in the polynomial category from start to finish. In particular, there is no hidden birational simplification that could alter the support rows or insert exceptional cancellation phenomena.

Lemma 3.1 (Canonicity). *The maps S and T are polynomial automorphisms of $\mathbb{A}^6$, and $F_g = T \circ S$ preserves the standard symplectic form.*

Row	Support vectors
$\partial_{q_1} V$	$(1, 2, 2, 0, 0, 0), (g-1, 0, 0, 0, 0, 0)$
$\partial_{q_2} V$	$(2, 1, 2, 0, 0, 0)$
$\partial_{q_3} V$	$(2, 2, 1, 0, 0, 0)$
$\partial_{p_1} W$	$(0, 0, 0, 1, 2, 2)$
$\partial_{p_2} W$	$(0, 0, 0, 2, 1, 2)$
$\partial_{p_3} W$	$(0, 0, 0, 2, 2, 1), (0, 0, 0, 0, 0, g-1)$

Table 4: The complete six-row support ledger. No omitted row is used in the proof.

Proof. The triangular inverses follow immediately from the explicit formulas above. Write the Jacobians in block form using the Hessians $H_V = \nabla^2 V(q)$ and $H_W = \nabla^2 W(P)$, where $P = p + \nabla V(q)$ is the intermediate p -coordinate after applying S :

$$J_S = \begin{pmatrix} I_3 & 0 \\ H_V & I_3 \end{pmatrix}, \quad J_T = \begin{pmatrix} I_3 & H_W \\ 0 & I_3 \end{pmatrix}.$$

Both Hessians are symmetric because they are Hessians of scalar polynomials. This symmetry is the only structural input needed for the symplectic check. With the standard symplectic matrix $\Omega = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}$, one first computes

$$\Omega J_S = \begin{pmatrix} H_V & I_3 \\ -I_3 & 0 \end{pmatrix}, \quad \Omega J_T = \begin{pmatrix} 0 & I_3 \\ -I_3 & -H_W \end{pmatrix}.$$

Multiplying on the left by the transposes gives

$$J_S^\top \Omega J_S = \begin{pmatrix} I_3 & H_V^\top \\ 0 & I_3 \end{pmatrix} \begin{pmatrix} H_V & I_3 \\ -I_3 & 0 \end{pmatrix} = \begin{pmatrix} H_V - H_V^\top & I_3 \\ -I_3 & 0 \end{pmatrix} = \Omega$$

and similarly

$$J_T^\top \Omega J_T = \begin{pmatrix} I_3 & 0 \\ H_W^\top & I_3 \end{pmatrix} \begin{pmatrix} 0 & I_3 \\ -I_3 & -H_W \end{pmatrix} = \begin{pmatrix} 0 & I_3 \\ -I_3 & H_W^\top - H_W \end{pmatrix} = \Omega.$$

Hence $J_{F_g} = J_T J_S$ is symplectic. The proof is independent of the later cone and selector arguments: canonicity is already forced by the gradient form of the two shears.

$$J_{F_g} = \begin{pmatrix} I_3 & H_W \\ 0 & I_3 \end{pmatrix} \begin{pmatrix} I_3 & 0 \\ H_V & I_3 \end{pmatrix} = \begin{pmatrix} I_3 + H_W H_V & H_W \\ H_V & I_3 \end{pmatrix}.$$

The product shows at a glance that the full map remains polynomial and that the upper-right block is exactly the Hessian block governing the second shear. It also gives $\det J_{F_g} = 1$, the determinant shadow of symplecticity. $\square$

3.2 Support ledger and the selected matrices

The support rows are recorded in the ordered basis $(q_1, q_2, q_3, p_1, p_2, p_3)$. They are the audit trail for every selected matrix entry.

Table 4 is complete in a literal sense. Each displayed row is the full support of one coordinate of one gradient. The q_2 - and q_3 -rows of ∇V have only one monomial each, and the p_1 - and p_2 -rows

Row	candidate score 1	candidate score 2	selector status
$\partial_{q_1} V$	$(g-1)u_1$	$u_1 + 2u_2 + 2u_3$	decided by M_S
$\partial_{q_2} V$	$2u_1 + u_2 + 2u_3$	—	rigid
$\partial_{q_3} V$	$2u_1 + 2u_2 + u_3$	—	rigid
$\partial_{p_1} W$	$v_1 + 2v_2 + 2v_3$	—	rigid
$\partial_{p_2} W$	$2v_1 + v_2 + 2v_3$	—	rigid
$\partial_{p_3} W$	$(g-1)v_3$	$2v_1 + 2v_2 + v_3$	decided by M_T

Table 5: Score ledger for the six support rows. The symbols u and $v = A_g u$ separate the first and second phases explicitly, so the two competitive rows are visible before the full-step matrix is invoked.

of ∇W also have only one monomial each. Consequently, the only genuine pure-versus-mixed competitions occur in the $\partial_{q_1} V$ row and the $\partial_{p_3} W$ row. That observation is the combinatorial reason that the selector analysis later reduces to exactly two faces.

The rigid singleton rows are not passive, however. They are exactly the rows that force the mixed coordinates of A_g and B_g to be positive in the pattern needed for C_g to become entrywise positive. In that sense the rows without a selector competition are what make the final recurrence matrix a primitive Perron object. There is also a complementary negative statement: no hidden support row remains offstage to create a third selector face after composition. Every exponent vector used later in the proof is already present in Table 4, and every weighted-degree comparison is a linear combination of those displayed rows.

The selected degree-transfer matrices are

$$A_g = \begin{pmatrix} g-1 & 0 & 0 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 0 & 0 & g-1 \end{pmatrix}.$$

The complete-step matrix is displayed separately to keep all three rows at full text width:

$$C_g = B_g A_g = \begin{pmatrix} g+7 & 6 & 6 \\ 2g+4 & 5 & 4 \\ 2(g-1) & 2(g-1) & g-1 \end{pmatrix}.$$

Every entry of A_g and B_g is read directly from Table 4. The first matrix records the selected q -to- p transfer and the second records the selected p -to- q transfer.

Reading Table 5 row by row clarifies why the mixed rows are not merely lower-order decoration. In the second and third S -rows the mixed monomials are the only monomials available, so they define the transfer matrix just as decisively as the pure q_1^{g-1} row. The same is true for the first two T -rows. The selector analysis is therefore not saying that mixed terms disappear; it is saying something narrower and more informative, namely that among the six displayed rows only two rows contain a genuine choice, and those two choices can be audited by explicit inequalities.

The full-step matrix C_g is not inserted by fiat. It is the literal product of the two phase matrices:

$$B_g A_g = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 0 & 0 & g-1 \end{pmatrix} \begin{pmatrix} g-1 & 0 & 0 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} = \begin{pmatrix} g+7 & 6 & 6 \\ 2g+4 & 5 & 4 \\ 2(g-1) & 2(g-1) & g-1 \end{pmatrix}.$$

Thus the first row of C_g is the sum of the first row of A_g plus twice the second and third rows of A_g ; the second and third rows are obtained in the same way. This explicit multiplication matters because

later inequalities are proved for C_g itself, not for an abstract matrix with the same characteristic polynomial. Table 5 makes one bookkeeping point explicit. The S -phase is evaluated against the u -weights, while the T -phase is evaluated against the intermediate weights $v = A_g u$. The notation prevents a common error in two-phase degree arguments, namely using the same weight vector on both sides of the phase boundary. Here the boundary is real: the second phase sees the support created by the first, not the original u -weights directly.

Lemma 3.2 (Reading the transfer matrices from the support ledger). *The six support rows in Table 4 determine the displayed matrices A_g and B_g uniquely.*

Proof. The S -phase reads the support rows of ∇V . In the $\partial_{q_1} V$ row the candidates are the mixed exponent $(1, 2, 2)$ and the pure exponent $(g-1, 0, 0)$; the selector analysis in the next section decides which one dominates. In the $\partial_{q_2} V$ and $\partial_{q_3} V$ rows there is no choice at all, because each row has a single mixed monomial. This yields the three rows of A_g :

$$(g-1, 0, 0), \quad (2, 1, 2), \quad (2, 2, 1).$$

The T -phase reads the support rows of ∇W . The $\partial_{p_1} W$ and $\partial_{p_2} W$ rows again have single mixed monomials, while the $\partial_{p_3} W$ row has a mixed candidate $(2, 2, 1)$ and a pure candidate $(0, 0, g-1)$. These are exactly the rows of B_g . No additional support row exists, so no other matrix can arise from the displayed family.

- The first S -row is the only row with a genuine pure-vs-mixed choice. It is the only place where a selector gap can appear before the cone is imposed.
- The second and third S -rows are rigid: their mixed monomials fix the second and third rows of A_g with no ambiguity.
- The first and second T -rows are equally rigid and therefore do not contribute selector faces.
- The third T -row is the only second-phase row that can compete with a pure term; that competition produces M_T .
- Because every row is accounted for explicitly, the support ledger is exhaustive rather than schematic.

This row-by-row reading is what makes the later degree induction honest: the matrices are not guessed, and no hidden monomial family is being suppressed. $\square$

Lemma 3.3 (Entrywise positivity of the selected matrices). *For $g \geq 8$, the matrix C_g is entrywise positive, and both A_g and B_g have nonnegative entries with positive row sums.*

Proof. The entries of A_g and B_g are visibly nonnegative from their definitions. Their row sums are $(g-1), 5, 5$ and $5, 5, g-1$ respectively, so every row is nonzero. The product matrix C_g has entries $g+7, 6, 6, 2g+4, 5, 4, 2(g-1), 2(g-1), g-1$, all of which are strictly positive for $g \geq 8$. This is the positivity input used later in the cone and Perron arguments. $\square$

Definition 3.4 (Weighted score and selected row). *For a positive vector $u = (u_1, u_2, u_3) \in \mathbb{R}_{>0}^3$ and an exponent vector $\alpha = (\alpha_1, \alpha_2, \alpha_3) \in \mathbb{N}^3$, define the weighted score*

$$\text{wt}_u(\alpha) = \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3.$$

Within any support row, the selected row is the monomial whose exponent vector has maximal weighted score.

This definition is the bridge from the support ledger to the matrix recurrence. Because multiplying u by a positive scalar multiplies every score by the same scalar, only the ratios u_2/u_1 and u_3/u_1 matter for selection. That scale-invariance is exactly why the selector problem can be reduced to the two normalized variables x and y . In the competitive S -row one obtains

$$\text{wt}_u(g-1, 0, 0) - \text{wt}_u(1, 2, 2) = u_1((g-2) - 2x - 2y) = u_1 M_S$$

whereas the competitive T -row must be evaluated at the intermediate weight $v = A_g u$, giving

$$\text{wt}_v(0, 0, g-1) - \text{wt}_v(2, 2, 1) = u_1((2g-6)x + (g-6)y - 6) = u_1 M_T.$$

Thus M_S and M_T are not ad hoc inequalities; they are precisely the normalized score gaps of the only two competitive support rows, with the phase change made explicit in the second comparison.

Lemma 3.5 (Explicit normalized selector derivation). *Let $u = (u_1, u_2, u_3) \in \mathbb{R}_{>0}^3$ and write $x = u_2/u_1$, $y = u_3/u_1$. Then the S - and T -selector gaps arise by direct score subtraction from the two competitive rows and from no others.*

Proof. In the competitive S -row one compares the mixed exponent $(1, 2, 2)$ with the pure exponent $(g-1, 0, 0)$. Their scores are

$$\text{wt}_u(1, 2, 2) = u_1 + 2u_2 + 2u_3 = u_1(1 + 2x + 2y)$$

and

$$\text{wt}_u(g-1, 0, 0) = (g-1)u_1.$$

Subtracting gives

$$\text{wt}_u(g-1, 0, 0) - \text{wt}_u(1, 2, 2) = u_1((g-2) - 2x - 2y) = u_1 M_S.$$

For the competitive T -row one must first pass through the selected S -phase vector $v = A_g u$, so

$$v_1 = (g-1)u_1, \quad v_2 = (2+x+2y)u_1, \quad v_3 = (2+2x+y)u_1.$$

The mixed T -candidate $(2, 2, 1)$ has score

$$2v_1 + 2v_2 + v_3 = u_1(2g + 4 + 4x + 5y),$$

while the pure T -candidate $(0, 0, g-1)$ has score

$$(g-1)v_3 = u_1((2g-2) + (2g-2)x + (g-1)y).$$

Subtracting yields

$$(g-1)v_3 - (2v_1 + 2v_2 + v_3) = u_1((2g-6)x + (g-6)y - 6) = u_1 M_T.$$

All other support rows are singletons, so they admit no nontrivial selector gap. $\square$

The second-phase computation is worth isolating conceptually. The formula for M_T is not read directly from the p -support row in isolation; it is read from that row after the first-phase selection has already replaced the original weights u by the intermediate weights $v = A_g u$. This is the first place where the two-phase character of the family becomes unavoidable. The asymmetry between M_S and M_T is therefore not an accident of notation. It records the exact point at which the first selector influences the second.

4 Selector gaps and cone invariance

4.1 Two and only two selector gaps

Let $u = (u_1, u_2, u_3)^\top$ be a positive weight vector and set $x = u_2/u_1$, $y = u_3/u_1$. The comparison of the pure row q_1^g with the mixed row $q_1^2 q_2^2 q_3^2$ gives

$$M_S = (g - 2) - 2x - 2y.$$

The comparison of the pure row p_3^g with the mixed row $p_1^2 p_2^2 p_3$ gives

$$M_T = (2g - 6)x + (g - 6)y - 6.$$

The normalization by u_1 is chosen so that the selector problem depends on ratios rather than on the absolute scale of the degree vector. Once $u_1 > 0$ is factored out, the mixed support comparison becomes a planar question in the (x, y) -variables. The lower bounds $x, y \geq 1$ encode the fact that the second and third coordinates should not fall below the first, while the upper bound $x + y < (g - 3)/2$ is exactly the margin needed for the pure q_1^g term to keep beating the mixed $q_1^2 q_2^2 q_3^2$ term. This is why the cone is written in normalized coordinates rather than as a homogeneous region in $\mathbb{R}_{>0}^3$.

The normalized cone also keeps the selector problem separate from the growth problem. The signs of M_S and M_T decide which support rows win inside the individual shears. By contrast, the positivity of C_g and the later Perron argument concern what happens only after those winners have already been frozen. Working in (x, y) -coordinates prevents a logical shortcut: a face is selected because its normalized score is larger, not because the eventual full-step matrix turns out to have positive entries.

Lemma 4.1 (Seed location). *The seed $u_0 = \mathbf{1}$ belongs to $\mathcal{K}_g$ for every $g \geq 8$. For $g = 7$, the same seed lies on the boundary $x + y = (g - 3)/2$.*

Proof. For $u_0 = \mathbf{1}$ one has $x = y = 1$. Thus $x, y \geq 1$ always. If $g \geq 8$, then

$$x + y = 2 < \frac{g - 3}{2},$$

so $u_0 \in \mathcal{K}_g$. If $g = 7$, then $x + y = 2 = (g - 3)/2$, so the seed lies on the boundary rather than in the open cone. $\square$

The ledger in Table 6 separates two different tasks. The quantities M_S and M_T decide which support row wins inside a fixed shear. The quantities $X' - 1$, $Y' - 1$, and H_2 belong to the full-step matrix and decide whether the normalized cone is preserved from one iterate to the next. Keeping those tasks separate prevents a common source of drift, namely using a full-step estimate to justify a half-step selector choice.

This separation also explains why the proof needs exactly two selector gaps and no more. A third gap would require a third row with two competing monomials, but the support ledger has only two such rows. All other rows are rigid, so they can change the value of a selected degree vector only after the selected support has already been decided. In other words, the phase structure is not an artifact of the proof; it is already present in the support ledger itself.

Lemma 4.2 (Exactly two selector faces). *For the displayed family there are exactly two nontrivial selector comparisons: M_S in the S -phase and M_T in the T -phase. No third selector face occurs.*

Ledger item	Formula	Role in the proof
S -selector gap	$M_S = (g - 2) - 2x - 2y$	pure q_1^g versus mixed $q_1^2 q_2^2 q_3^2$ in the first shear
T -selector gap	$M_T = (2g - 6)x + (g - 6)y - 6$	pure p_3^g versus mixed $p_1^2 p_2^2 p_3$ in the second shear
$X' - 1$ test	$\frac{g - 3 - x - 2y}{g + 7 + 6x + 6y}$	proves the lower bound $X' > 1$ on the normalized cone
$Y' - 1$ test	$\frac{g - 9 + (2g - 8)x + (g - 7)y}{g + 7 + 6x + 6y}$	proves the lower bound $Y' > 1$
height test	$H_2 = (g - 3)U'_1 - 2U'_2 - 2U'_3$	proves $x' + y' < (g - 3)/2$ after one full step

Table 6: Selector and cone ledger. Exactly two selector faces appear; the remaining quantities are normalized cone tests for the full-step matrix C_g .

Proof. By Table 4, the S -phase has three rows to inspect. The $\partial_{q_2} V$ and $\partial_{q_3} V$ rows each contain a single monomial, so they cannot produce a selector boundary. Only the $\partial_{q_1} V$ row contains both a mixed and a pure candidate, and their weight difference is M_S . The same reasoning applies to the T -phase: the $\partial_{p_1} W$ and $\partial_{p_2} W$ rows are single-support rows, while $\partial_{p_3} W$ contains the unique pure-versus-mixed choice with gap M_T . Since there are no further support rows, there is no third selector face. $\square$

Lemma 4.3 (Selector positivity). *For $g \geq 8$ and (x, y) in the cone $\mathcal{K}_g$, both M_S and M_T are strictly positive.*

Proof. On $\mathcal{K}_g$, the height condition $x + y < (g - 3)/2$ implies

$$M_S = (g - 2) - 2x - 2y > (g - 2) - 2 \cdot \frac{g - 3}{2} = 1.$$

Since $x, y \geq 1$,

$$M_T = (2g - 6)x + (g - 6)y - 6 \geq (2g - 6) + (g - 6) - 6 = 3g - 18 \geq 6.$$

Hence both gaps are strict. The case $g = 7$ is not inside the open cone and is recorded separately as a boundary note. $\square$

4.2 Cone invariance

Let $U' = C_g U$. With $x = U_2/U_1$, $y = U_3/U_1$, and $D = g + 7 + 6x + 6y$, direct substitution yields

$$U'_1 = DU_1, \quad U'_2 = (2g + 4 + 5x + 4y)U_1, \quad U'_3 = (2(g - 1) + 2(g - 1)x + (g - 1)y)U_1.$$

Hence the normalized coordinates $X' = U'_2/U'_1$ and $Y' = U'_3/U'_1$ satisfy

$$X' - 1 = \frac{g - 3 - x - 2y}{D}, \quad Y' - 1 = \frac{g - 9 + (2g - 8)x + (g - 7)y}{D}.$$

Lemma 4.4 (Facial slack ledger). *Put*

$$h = g - 3 - 2x - 2y.$$

On $\mathcal{K}_g$, one has $h > 0$, and the first selector margin and the first normalized cone margin admit the exact rewritings

$$M_S = 1 + h, \quad X' - 1 = \frac{x + h}{D}.$$

Moreover,

$$M_T \geq 3g - 18 \geq 6, \quad g - 9 + (2g - 8)x + (g - 7)y \geq 4g - 24 \geq 8.$$

Thus both selector decisions and both normalized lower faces are strict before the upper-height face is considered.

Proof. The definition of the open cone is precisely $h > 0$ together with $x, y \geq 1$. Substitution gives

$$M_S = (g - 2) - 2x - 2y = 1 + h$$

and

$$g - 3 - x - 2y = x + h,$$

which proves the two identities. For the second selector, the coefficients of x and y are positive when $g \geq 8$, so their minimum on the lower faces occurs at $x = y = 1$:

$$M_T \geq (2g - 6) + (g - 6) - 6 = 3g - 18.$$

The numerator of $Y' - 1$ is treated in the same way:

$$g - 9 + (2g - 8)x + (g - 7)y \geq g - 9 + (2g - 8) + (g - 7) = 4g - 24.$$

The final numerical lower bounds follow from $g \geq 8$. Notice that this is an exact facial calculation, not a compactness argument: the strictness of the first two target faces is visible directly in the four displayed margins. $\square$

The first two cone inequalities are therefore reduced to sign checks on two explicit numerators. The third cone inequality is encoded by the height functional $H = RU'_1 - U'_2 - U'_3$ with $R = (g - 3)/2$, or equivalently by $H_2 = 2H = (g - 3)U'_1 - 2U'_2 - 2U'_3$. These three tests are sufficient because the cone has exactly three defining inequalities after normalization: $x \geq 1$, $y \geq 1$, and $x + y < R$. Thus $X' - 1$ certifies the first lower bound, $Y' - 1$ certifies the second, and $H_2 > 0$ certifies the upper-height bound. No extra hidden face remains to be checked once these three signs are positive. To expose every coefficient in the remaining test, substitute the three rows of C_g before collecting terms:

$$\begin{aligned} \frac{H_2}{U_1} &= (g - 3)(g + 7 + 6x + 6y) \\ &\quad - 2(2g + 4 + 5x + 4y) - 2(2(g - 1) + 2(g - 1)x + (g - 1)y) \\ &= g^2 - 4g - 25 + 2(g - 12)x + 4(g - 6)y. \end{aligned}$$

Consequently

$$H_2 = (g - 3)U'_1 - 2U'_2 - 2U'_3 = U_1Q,$$

where

$$Q = g^2 + 2gx + 4gy - 4g - 24x - 24y - 25, \quad H = \frac{H_2}{2} = RU'_1 - U'_2 - U'_3.$$

The lower-height parameter is $R = (g - 3)/2$. The split $8 \leq g \leq 11$ versus $g \geq 12$ is the only place where the inequality needs a case distinction. For the smaller range the coefficient of x in Q is negative, so one uses the cone inequality $x < (g - 3)/2 - y$ to eliminate x :

$$\begin{aligned} Q &= g^2 - 4g - 25 + 2(g - 12)x + 4(g - 6)y \\ &> g^2 - 4g - 25 + 2(g - 12)\left(\frac{g - 3}{2} - y\right) + 4(g - 6)y \\ &= 2g^2 - 19g + 11 + 2gy \\ &\geq 2g^2 - 17g + 11. \end{aligned}$$

This yields

$$Q > 2g^2 - 17g + 11 \geq 3 \quad (8 \leq g \leq 11),$$

because the last comparison has the exact factorization

$$2g^2 - 17g + 11 = 3 + (g - 8)(2g - 1).$$

For the larger range all coefficients of x and y are nonnegative, so the trivial lower bounds $x, y \geq 1$ are enough:

$$\begin{aligned} Q &= g^2 - 4g - 25 + 2(g - 12)x + 4(g - 6)y \\ &\geq g^2 - 4g - 25 + 2(g - 12) + 4(g - 6) \\ &= g^2 + 2g - 73. \end{aligned}$$

Thus

$$Q \geq g^2 + 2g - 73 > 0 \quad (g \geq 12).$$

Here the lower endpoint is also explicit:

$$g^2 + 2g - 73 = 95 + (g - 12)(g + 14),$$

so the large- g estimate has a positive margin already at $g = 12$ and only increases afterwards. No numerical trial is hidden here; the split is forced only by the sign change of the x -coefficient in Q . Therefore $H_2 > 0$, hence $H > 0$, and so $x' + y' < (g - 3)/2$.

The geometry behind the split is simple but essential. For $8 \leq g \leq 11$, the coefficient $2(g - 12)$ is negative, so increasing x makes the height test worse rather than better. In that range the sharp move is to push x to the boundary value $(g - 3)/2 - y$ supplied by the cone and then use only the lower bound $y \geq 1$. For $g \geq 12$, the sign reverses and both x and y become helpful directions, so the coarse lower bounds $x, y \geq 1$ already suffice. The proof therefore uses the weakest valid estimate in each range and never appeals to a sampled picture of the cone.

Proposition 4.5 (Cone invariance). *If $U \in \mathcal{K}_g$, then $U' = C_g U \in \mathcal{K}_g$.*

Proof. Because $D > 0$, the sign of $X' - 1$ is the sign of $g - 3 - x - 2y$. On the cone one has $x + 2y < (g - 3)/2 + y \leq g - 4 < g - 3$, so $X' - 1 > 0$. Likewise $Y' - 1 > 0$ because $g - 9 \geq -1$ and the coefficients of x and y are nonnegative, with at least one strictly positive contribution once $x, y \geq 1$. The height inequality follows from the already verified positivity of $H_2 = U_1 Q$. Hence $X', Y' \geq 1$ and $X' + Y' < (g - 3)/2$, which is exactly the condition $U' \in \mathcal{K}_g$. $\square$

Range	sign pattern in Q	bound used
$8 \leq g \leq 11$	coefficient of x is negative, coefficient of y is positive	substitute $x < (g - 3)/2 - y$ and then use $y \geq 1$
$g \geq 12$	coefficients of x and y are nonnegative	substitute $x, y \geq 1$ directly

Table 7: Exact sign split in the cone proof. The two cases are the only ones needed because the sign of $2(g - 12)$ changes at $g = 12$.

The force of Proposition 4.5 is that it returns the iterate to the *open* cone and not merely to its closure. That distinction is what lets the selector proof restart with strict inequalities at every step. A closed cone argument would leave open the possibility of landing on a boundary where a future selector tie could occur. The present proof needs the stronger conclusion: once inside the cone, every full iterate comes back with room to spare, so the same two selector winners remain available indefinitely.

Remark 4.6 (Boundary case $g = 7$). At $g = 7$, the seed $(x, y) = (1, 1)$ lies on the boundary $x + y = (g - 3)/2$, and the first gap is $M_S = 1$. The seed is therefore a strict-cone boundary example, not a selector tie.

5 Phase/carry induction and characteristic-zero no-cancellation

5.1 Phase vectors

Let $u_0 = \mathbf{1} = (1, 1, 1)^\top$, and let v_{n+1} and u_{n+1} be the selected p - and q -degree vectors after the first and second shears at time n . The phase convention is

$$v_{n+1} = A_g u_n, \quad u_{n+1} = C_g u_n.$$

The first equality reads the first selected face, and the second equality reads the complete step. The distinction matters throughout the section. The intermediate vector v_{n+1} is the selected support after applying only S , while u_{n+1} is the selected support after the full map $T \circ S$. The proof keeps those two phases separate so that the half-step matrix A_g is never confused with the full-step matrix C_g .

There are two kinds of bookkeeping to maintain at once. One is geometric: which face wins inside a given support row. The other is temporal: whether the newly created term at time n is still larger than the term carried from time $n - 1$. The selector gaps M_S and M_T resolve the first issue, while Lemma 5.1 resolves the second by showing that the full-step degree vector grows strictly in every coordinate.

The iterate structure is easiest to read as a chain:

$$u_0 \xrightarrow{S} v_1 \xrightarrow{T} u_1 \xrightarrow{S} v_2 \xrightarrow{T} u_2 \xrightarrow{S} \dots$$

At each S -arrow the first matrix A_g acts, and at each T -arrow the second matrix B_g acts on the selected support to produce the full-step matrix C_g . The point of the chain is not decorative; it prevents the common mistake of comparing a carried term from time $n - 1$ directly against a full-step term at time $n + 1$ without accounting for the intermediate phase.

The first symbolic iterate already exhibits the later mechanism in miniature:

$$v_1 = A_g \mathbf{1} = (g - 1, 5, 5)^\top, \quad u_1 = C_g \mathbf{1} = (g + 19, 2g + 13, 5(g - 1))^\top.$$

The first vector shows the immediate S -phase transfer, while the second already contains the three row sums that later reappear in the Perron bound. It also shows that the full-step third coordinate is largest at the seed itself, since $5(g-1) > 2g+13 > g+19$ for $g \geq 8$. The induction below should therefore be read as a persistence argument for a pattern that is visible from the first full iterate and not as a wholly different asymptotic phenomenon.

One more exact step already displays the quadratic growth pattern hidden behind the linear recurrence:

$$\begin{aligned} v_2 &= A_g u_1 = ((g-1)(g+19), 14g+41, 11g+59)^\top, \\ u_2 &= C_g u_1 = (g^2+68g+181, 2g^2+72g+121, 11g^2+48g-59)^\top. \end{aligned}$$

These formulas are not needed for the abstract induction, but they provide a useful audit of what the induction is claiming. The full-step third coordinate already remains visible at the second iterate because

$$u_{2,3} - u_{2,2} = 3(g-6)(3g+10) > 0, \quad u_{2,3} - u_{2,1} = 10(g-6)(g+4) > 0$$

for every $g \geq 8$. In other words, the visible-row phenomenon is not only eventually true; it is already visible in the first two exact full iterates, and the later cone argument explains why this symbolic dominance persists for all subsequent iterates.

5.2 The inductive state in all six coordinates

Write the n -th iterate in coordinates as

$$F_g^n(q, p) = (Q^{(n)}(q, p), P^{(n)}(q, p)),$$

where $Q^{(n)} = (Q_1^{(n)}, Q_2^{(n)}, Q_3^{(n)})$ and similarly for $P^{(n)}$. Applying one more complete step first forms

$$\tilde{P}^{(n+1)} = P^{(n)} + \nabla V(Q^{(n)})$$

and then forms

$$Q^{(n+1)} = Q^{(n)} + \nabla W(\tilde{P}^{(n+1)}), \quad P^{(n+1)} = \tilde{P}^{(n+1)}.$$

These identities show exactly where the two carried vectors enter. The old $P^{(n)}$ competes with the fresh V -gradient during the first phase; the old $Q^{(n)}$ competes with the fresh W -gradient during the second. Thus a proof that checks only the internal pure-versus-mixed selectors is not yet an induction: the carried coordinate in each displayed sum must also lose strictly.

Let

$$\mathbf{d}_q^{(n)} = (\deg Q_1^{(n)}, \deg Q_2^{(n)}, \deg Q_3^{(n)})^\top, \quad \mathbf{d}_p^{(n)} = (\deg P_1^{(n)}, \deg P_2^{(n)}, \deg P_3^{(n)})^\top.$$

The induction will certify the six-coordinate state

$$\mathbf{d}_q^{(n)} = u_n, \quad \mathbf{d}_p^{(n)} = v_n \quad (n \geq 1),$$

with $\mathbf{d}_q^{(0)} = \mathbf{d}_p^{(0)} = \mathbf{1}$. This statement is not assumed in defining u_n, v_n : it is the conclusion obtained after strict selection, carried-term dominance, and coefficient survival have all been checked. The two phases can be summarized as follows.

Table 8 makes the indexing transparent. The vector v_{n+1} is both the output of the new S -phase and the p -degree vector retained at the end of the complete step. The vector u_{n+1} is created only after the T -phase. In particular, $B_g v_{n+1}$ may be replaced by $C_g u_n$ only after the first equality in the table has been proved. This is the phase-carry dependency that the remaining lemmas close.

Phase	carried vector	fresh selected vector	exact output after comparison
S	v_n for $n \geq 1$, and $\mathbf{1}$ at the seed	$A_g u_n$	$v_{n+1} = A_g u_n$
T	u_n	$B_g v_{n+1} = C_g u_n$	$u_{n+1} = C_g u_n$

Table 8: Six-coordinate induction state. An internal selector chooses the fresh row; a separate strict comparison then removes the carried coordinate from contention.

Lemma 5.1 (Strict coordinate growth). *For $g \geq 8$, the matrix*

$$C_g - I = \begin{pmatrix} g+6 & 6 & 6 \\ 2g+4 & 4 & 4 \\ 2g-2 & 2g-2 & g-2 \end{pmatrix}$$

is entrywise positive. Consequently, whenever $u_{n+1} = C_g u_n$ with $u_n \in \mathbb{R}_{>0}^3$, one has $u_{n+1} > u_n$ coordinatewise.

Proof. Every entry of $C_g - I$ is strictly positive for $g \geq 8$. Therefore $(C_g - I)u_n > 0$ for every positive vector u_n , which is equivalent to $u_{n+1} - u_n > 0$. $\square$

Lemma 5.2 (Carried q -terms in the T -phase). *Suppose that the S -phase has selected $v_{n+1} = A_g u_n$. Then the fresh T -gradient degree vector strictly dominates the carried q -degree vector, including at the base step:*

$$B_g v_{n+1} - u_n = C_g u_n - u_n = (C_g - I)u_n > 0, \quad n \geq 0.$$

Consequently the carried coordinates q_1, q_2, q_3 cannot alter the selected full-step vector $u_{n+1} = C_g u_n$.

Proof. The identity $B_g v_{n+1} = B_g A_g u_n = C_g u_n$ is the phase product. Because every u_n is positive and $C_g - I$ is entrywise positive by Lemma 5.1, the displayed difference is positive in all three coordinates. For $n = 0$ this reads $(C_g - I)\mathbf{1} > 0$, so the argument includes the initially carried coordinate functions of degree one rather than starting only after the first iterate. The selector gap M_T has a separate role: it chooses the pure term over the mixed term inside the competitive third W -row. The present lemma is what compares those freshly selected W -gradient rows with the three carried q -coordinates. $\square$

Proposition 5.3 (Base step). *At $n = 0$, the selected S -face dominates the carried term, and $v_1 = A_g u_0 > \mathbf{1}$.*

Proof. Lemma 4.1 places the seed in the cone when $g \geq 8$. Lemma 4.3 then gives $M_S > 0$, so the pure-versus-mixed comparison on the S -phase is strict. Since

$$A_g \mathbf{1} = (g-1, 5, 5)^\top,$$

the selected gradient terms strictly dominate the carried p -terms. At the base step the carried p -coordinates are simply the original coordinate functions p_1, p_2, p_3 , all of degree one, so there is no hidden competitor with the same weight. $\square$

Table 9 is a useful audit because it shows the whole mechanism once before the induction begins. The seed already lies in the cone, the first S -phase applies the pure q_1^{g-1} row and the two rigid mixed rows, and the first full-step vector exhibits the three row sums of C_g . In particular,

$$\frac{2g+13}{g+19} > 1, \quad \frac{5(g-1)}{g+19} > 1,$$

Stage	exact vector
seed u_0	$(1, 1, 1)^\top$
first S -phase v_1	$(g - 1, 5, 5)^\top$
first full step u_1	$(g + 19, 2g + 13, 5(g - 1))^\top$

Table 9: First-step audit of the phase-labelled recurrence. The seed, the first S -phase vector, and the first full-step vector are all explicit.

Carried row	fresh-minus-carried margin	sign reason
first p -row	$(g - 1)\Delta_{n,1}$	$\Delta_{n,1} > 0$
second p -row	$2\Delta_{n,1} + \Delta_{n,2} + 2\Delta_{n,3}$	positive coefficient sum
third p -row	$2\Delta_{n,1} + 2\Delta_{n,2} + \Delta_{n,3}$	positive coefficient sum

Table 10: Coordinatewise carried-term margins in the S -phase, where $\Delta_n = u_n - u_{n-1}$. Every carried p -coordinate is beaten row by row, not merely in aggregate.

and

$$\frac{2g + 13 + 5(g - 1)}{g + 19} = \frac{7g + 8}{g + 19} < \frac{g - 3}{2}$$

because $g^2 + 2g - 73 > 0$ for $g \geq 8$. So the first full iterate is not merely positive; it already satisfies the cone inequalities with the same symbolic margin pattern used later in the general proof. The induction does not create a new phenomenon after the base step. It formalizes the persistence of the structure already visible in the first exact iterate.

Proposition 5.4 (Old-term induction). *For every $n \geq 1$, if $p_n = A_g u_{n-1}$, then the S -selector remains strict, the intermediate vector satisfies $A_g u_n > p_n$, and the T -phase selects $u_{n+1} = C_g u_n$.*

Proof. Assume $p_n = A_g u_{n-1}$. By Proposition 4.5, every u_n lies in $\mathcal{K}_g$, and by Lemma 5.1 one has $u_n > u_{n-1}$ coordinatewise. Since A_g has nonnegative entries and each row has at least one strictly positive entry, it follows that

$$A_g u_n > A_g u_{n-1} = p_n.$$

Writing $\Delta_n = u_n - u_{n-1}$, the strict difference is

$$A_g \Delta_n = \begin{pmatrix} (g - 1)\Delta_{n,1} \\ 2\Delta_{n,1} + \Delta_{n,2} + 2\Delta_{n,3} \\ 2\Delta_{n,1} + 2\Delta_{n,2} + \Delta_{n,3} \end{pmatrix} > 0.$$

This is the old-term dominance statement: the newly selected S -term is strictly larger than the carried p -term from the previous iterate. The T -selector is then controlled by the second explicit gap M_T , whose strict positivity was proved in Lemma 4.3. After that internal face comparison, Lemma 5.2 proves separately that $C_g u_n - u_n > 0$, so all three freshly selected T -gradient rows beat the carried q -coordinates. Thus the same two faces are selected again and the full step is exactly $u_{n+1} = C_g u_n$. At no stage do we invoke a preservation of arbitrary face hyperplanes; we use the two explicit gaps M_S and M_T , the S -carry margins, and the explicit T -carry margin $(C_g - I)u_n$. $\square$

Table 10 is the temporal counterpart of the selector ledger. The selector gaps say which exponent vector would win if one compared only the new candidate monomials. The carry table says that

even after the older p -coordinate is brought back into the comparison, each row still wins by an explicit positive margin. This is stronger than a statement about total degree alone. It rules out the possibility that a row grows in aggregate while still losing to a carried term in one coordinate.

Lemma 5.5 (Unique selected exponent at each phase). *For every $n \geq 0$, each support row contributing to the recurrence has a unique selected exponent vector.*

Proof. The rigid rows in Table 4 are singletons, so uniqueness is automatic there. In the S -competitive row, Lemma 4.3 gives $M_S > 0$ on $\mathcal{K}_g$, so the pure exponent $(g-1, 0, 0)$ strictly beats the mixed exponent $(1, 2, 2)$. In the T -competitive row, the same lemma gives $M_T > 0$, so the pure exponent $(0, 0, g-1)$ strictly beats the mixed exponent $(2, 2, 1)$. Because the cone is invariant, these strict inequalities are available at every iterate. Hence each selected row in the recurrence has a unique winner before any coefficient bookkeeping is considered. $\square$

5.3 No-cancellation in characteristic zero

Lemma 5.6 (Positive-support no-cancellation). *Let $R = \mathbb{N}[z_1, \dots, z_6]$. If a polynomial is written as a finite sum of distinct monomials with positive coefficients, then substituting polynomials from R preserves positivity of every selected Minkowski-sum coefficient. Over a field of characteristic zero, the resulting selected coefficient is nonzero.*

Proof. Distributivity expands the substituted polynomial as a finite sum of products of positive integers. No sign changes occur, and characteristic zero prevents the positive integer coefficient from vanishing. In the present application the selected monomial support is obtained by Minkowski-summing already selected supports from the previous phase. Every contribution to the selected coefficient is therefore a positive integer, so the coefficient after substitution is a positive integer as well. The lemma is exactly the support-level version of the carry induction above: once the selector gaps are strict, the only remaining obstruction would be cancellation, and that obstruction is absent in characteristic zero.

The role of Lemma 5.5 is precisely to make this coefficient statement meaningful. Strict selector gaps identify a unique exponent vector in each competitive row, and the semiring argument then shows that the coefficient attached to that unique exponent remains positive after substitution. The proof does not rely on a general theorem about all substitutions of positive polynomials; it uses the much narrower fact that the selected exponent is unique and every contributing coefficient path is a positive integer path.

A representative example is the pure T -row. Once the selected S -phase support for p_3 is fixed, the pure monomial p_3^{g-1} expands into a product of $g-1$ identical positive-support polynomials. Every path contributing to the selected exponent adds a positive integer coefficient, and no negative sign is ever introduced. The same mechanism applies to the mixed rows, where the selected exponent is obtained by a Minkowski sum of already selected supports. Thus the no-cancellation lemma is not an abstract semiring tagline; it is the coefficient-level reason the exact selected row survives the entire iteration. $\square$

Lemma 5.7 (Explicit positive coefficient witnesses). *Assume the selected monomial in each $Q_i^{(n)}$ has a positive integer coefficient a_i . Then the three selected S -gradient monomials receive positive contributions with coefficients*

$$ga_1^{g-1}, \quad 2a_1^2a_2a_3^2, \quad 2a_1^2a_2^2a_3.$$

If the corresponding selected monomials of $\tilde{P}^{(n+1)}$ have positive coefficients b_1, b_2, b_3 , then the three selected T -gradient monomials receive positive contributions with coefficients

$$2b_1b_2^2b_3^2, \quad 2b_1^2b_2b_3^2, \quad gb_3^{g-1}.$$

Every one of these witnesses is nonzero in characteristic zero.

Proof. The first list is read directly from

$$\partial_{q_1} V = 2q_1q_2^2q_3^2 + gq_1^{g-1}, \quad \partial_{q_2} V = 2q_1^2q_2q_3^2, \quad \partial_{q_3} V = 2q_1^2q_2^2q_3.$$

The strict S -selector chooses the pure term in the first row, while the other two rows are singletons. Choosing one already selected monomial from each factor in the indicated products gives exactly the three displayed positive contributions. Likewise,

$$\partial_{p_1} W = 2p_1p_2^2p_3^2, \quad \partial_{p_2} W = 2p_1^2p_2p_3^2, \quad \partial_{p_3} W = 2p_1^2p_2^2p_3 + gp_3^{g-1}.$$

The first two rows are singletons and the strict T -selector chooses the pure term in the third row, producing the second list.

Other products may contribute to the same exponent, so the displayed numbers are coefficient witnesses rather than claims that no other path exists. That distinction is harmless and useful: all forward coordinate polynomials lie in the positive semiring $\mathbb{N}[q_1, q_2, q_3, p_1, p_2, p_3]$, so any additional path only adds another nonnegative integer. Since 2 and the positive integer g remain nonzero in characteristic zero, each selected exponent survives with a nonzero coefficient. The lemma therefore turns the abstract semiring statement into six explicit recursive witnesses, one for each fresh gradient row. $\square$

Theorem 5.8 (Exact phase recurrence). *The complete selected recurrence is*

$$v_{n+1} = A_g u_n, \quad u_{n+1} = C_g u_n \quad (n \geq 0),$$

and the selected support coefficient never cancels.

Proof. Combine Proposition 5.3, Lemma 5.2, Proposition 5.4, Lemma 5.5, and Lemmas 5.6–5.7. The recurrence is phase-labelled: the first matrix is the first shear, the second matrix is the complete step, and the support coefficient remains positive at every iteration. Both carried-coordinate comparisons are explicit: $A_g u_n$ beats the carried p -vector, while $C_g u_n$ beats the carried q -vector. In particular, the induction does not merely identify the maximal degree numerically; it identifies a specific support row whose coefficient remains nonzero through the entire iteration process.

This is the point at which characteristic zero is genuinely used. If the field had positive characteristic, a positive integer coefficient could vanish after substitution, and the support-level argument would no longer imply that the selected row survives. The paper therefore keeps the field hypothesis in the main theorem rather than hiding it in a remark, because it is part of the mechanism that makes the exact recurrence possible.

The conclusion of the theorem is stronger than a bare degree recurrence. It records a persistent support mechanism. At each stage one can point to the same selector rows, the same carried-coordinate comparisons, and the same positivity argument on coefficients. This is why the later visibility proof can safely read the degree from one coordinate functional: the recurrence is not an accidental equality of numbers but the shadow of a support pattern that has been preserved through every phase of the iteration. $\square$

6 Visibility, exact degree, and Perron bound

6.1 Visibility of the third coordinate

The matrix difference is

$$C_g - A_g = \begin{pmatrix} 8 & 6 & 6 \\ 2g+2 & 4 & 2 \\ 2g-4 & 2g-4 & g-2 \end{pmatrix},$$

which is entrywise positive for $g \geq 8$. Furthermore,

$$U_3 - U_2 = [-6 + (2g-7)x + (g-5)y]U_1 > 0, \quad U_3 - U_1 = [g-9 + (2g-8)x + (g-7)y]U_1 > 0.$$

The entrywise positivity of $C_g - A_g$ controls the q - versus p -phase comparison, while the two displayed differences control the internal ordering of the q -coordinates. Together they show that the visible coordinate is not merely one of several maximal rows: it is the unique row that dominates the other five coordinate degrees after each full iterate. Hence q_3 is the maximal coordinate degree among all six coordinates.

Lemma 6.1 (Six-coordinate dominance). *For every $n \geq 1$, the full-step degree vectors satisfy*

$$u_n > v_n$$

coordinatewise, and the third component of u_n is strictly larger than the first two components. Consequently the third q -coordinate has degree strictly larger than each of the other five coordinate functions of F_g^n .

Proof. Write the input to the n -th full step as $w = u_{n-1} = (W_1, W_2, W_3)^\top$, with $x = W_2/W_1$ and $y = W_3/W_1$. The exact recurrence gives

$$u_n - v_n = (C_g - A_g)w > 0$$

because $C_g - A_g$ is entrywise positive. This compares corresponding q_i - and p_i -rows.

For the internal q -ordering, put $u_n = (U_1, U_2, U_3)^\top$. Subtracting rows of C_g gives

$$\begin{aligned} U_3 - U_2 &= [-6 + (2g-7)x + (g-5)y]W_1, \\ U_3 - U_1 &= [g-9 + (2g-8)x + (g-7)y]W_1. \end{aligned}$$

Since $x, y \geq 1$, their bracketed coefficients have the explicit lower bounds

$$-6 + (2g-7) + (g-5) = 3g-18 \geq 6, \quad g-9 + (2g-8) + (g-7) = 4g-24 \geq 8.$$

Thus $U_3 > U_2, U_1$. Finally, the dominance chains

$$U_3 > U_1 > \deg P_1^{(n)}, \quad U_3 > U_2 > \deg P_2^{(n)}, \quad U_3 > \deg P_3^{(n)}$$

compare the visible row with all three p -coordinates. Hence it is the unique maximal degree among all six coordinate functions. $\square$

Writing $u_{n-1} = (U_1, U_2, U_3)^\top$ with $x = U_2/U_1$ and $y = U_3/U_1$, the phase difference is

$$u_n - v_n = \begin{pmatrix} (8 + 6x + 6y)U_1 \\ (2g+2 + 4x + 2y)U_1 \\ (2g-4 + (2g-4)x + (g-2)y)U_1 \end{pmatrix},$$

Comparison to certify	Explicit witness
q_i beats corresponding p_i after a full step	the i -th row of $(C_g - A_g)u_{n-1}$ is strictly positive
q_3 beats q_2	$U_3 - U_2 = [-6 + (2g - 7)x + (g - 5)y]U_1 > 0$
q_3 beats q_1	$U_3 - U_1 = [g - 9 + (2g - 8)x + (g - 7)y]U_1 > 0$
visible degree row	combine the previous two lines with characteristic-zero no-cancellation

Table 11: Visibility ledger. The exact degree row is fixed only after both the phase comparison and the internal q -ordering have been certified.

First full-step comparison	exact margin at $u_0 = \mathbf{1}$
$u_1 - v_1$, first coordinate	20
$u_1 - v_1$, second coordinate	$2g + 8$
$u_1 - v_1$, third coordinate	$5g - 10$
$u_{1,3} - u_{1,2}$	$3g - 18$
$u_{1,3} - u_{1,1}$	$4g - 24$

Table 12: Exact first-step visibility margins. The visible row is already strict at the first full iterate, and the cone proof shows that these margins persist symbolically thereafter.

so every full-step q -coordinate dominates the corresponding intermediate p -coordinate by an explicit positive margin. The third row then dominates the first two rows because the two displayed inequalities show $U_3 - U_2 > 0$ and $U_3 - U_1 > 0$. The visible row is therefore fixed by two independent comparisons: one across the phase boundary and one within the q -phase. Table 11 records these comparisons in the order they are actually used. The proof does not maximize over all six coordinates at once. First it reduces the question from six rows to the three full-step q -coordinates by using $C_g - A_g > 0$. Only after that reduction does it compare the three q -rows among themselves. This order matters because the functional $e_3^\top$ is an output of the visibility proof and not an assumption built into the recurrence from the beginning.

Proposition 6.2 (Exact degree row). *For every $n \geq 0$,*

$$\deg(F_g^n) = e_3^\top C_g^n \mathbf{1}.$$

Proof. The case $n = 0$ is immediate: $\deg(F_g^0) = \deg(\text{id}) = 1 = e_3^\top \mathbf{1}$. For $n \geq 1$, the phase recurrence gives the selected q -degree vector $u_n = C_g^n \mathbf{1}$ and the intermediate p -degree vector $v_n = A_g u_{n-1}$. Because

$$u_n - v_n = (C_g - A_g)u_{n-1},$$

and $C_g - A_g$ is entrywise positive for $g \geq 8$, every q -coordinate degree after the full step exceeds the corresponding intermediate p -coordinate degree. Among the q -coordinates, the displayed inequalities give $U_3 > U_2$ and $U_3 > U_1$. Since the visible coefficient never cancels, the degree of F_g^n is read off by the third-coordinate functional and equals $e_3^\top C_g^n \mathbf{1}$. $\square$

Table 12 shows that the degree identity is not a late asymptotic coincidence. Already at the first full iterate, every q -coordinate beats the corresponding p -coordinate and the third q -coordinate beats the other two q -coordinates by explicit positive amounts. The later cone and induction arguments upgrade this first-step audit into a uniform statement for all n : once the iterate stays in the cone, the same symbolic margins continue to separate the visible row from its competitors. This is why

the degree functional can be written as the fixed row vector $e_3^\top$ rather than as a changing “take the maximum” operation.

One can read the first values of the exact degree sequence directly from this audit:

$$\deg(F_g^0) = 1, \quad \deg(F_g^1) = 5(g-1), \quad \deg(F_g^2) = 11g^2 + 48g - 59.$$

These values are not obtained by guessing a scalar recurrence first and then verifying it afterward. They are obtained by following the visible row of the matrix recurrence that has already been certified symbolically. This point becomes important in the next section: once the characteristic polynomial of C_g is known, it governs the *exact* degree sequence because the degree has already been tied to one fixed row of C_g^n , not merely bounded above by it.

Corollary 6.3 (Perron bound). *The matrix C_g is primitive, hence*

$$\lambda_1(F_g) = \rho(C_g).$$

Its row sums are $g+19$, $2g+13$, and $5(g-1)$, all strictly below $(g-1)^2$ for $g \geq 8$. Therefore

$$\rho(C_g) < (g-1)^2.$$

Proof. C_g itself is entrywise positive for $g \geq 8$, hence primitive. Perron–Frobenius therefore provides a positive right eigenvector $r \in \mathbb{R}_{>0}^3$ for the simple eigenvalue $\rho(C_g)$. Choose positive constants a, b such that

$$ar \leq \mathbf{1} \leq br$$

coordinatewise; for example, take $a = \min_i r_i^{-1}$ and $b = \max_i r_i^{-1}$. Positivity of C_g preserves these inequalities, so

$$a\rho(C_g)^n r_3 \leq e_3^\top C_g^n \mathbf{1} \leq b\rho(C_g)^n r_3.$$

Taking n -th roots proves directly that

$$\lim_{n \rightarrow \infty} (e_3^\top C_g^n \mathbf{1})^{1/n} = \rho(C_g).$$

By Proposition 6.2,

$$\deg(F_g^n) = e_3^\top C_g^n \mathbf{1},$$

so the dynamical degree $\lambda_1(F_g) = \lim_{n \rightarrow \infty} \deg(F_g^n)^{1/n}$ is exactly the Perron root of C_g . The row sums are

$$(g+7) + 6 + 6 = g+19, \quad (2g+4) + 5 + 4 = 2g+13, \quad 2(g-1) + 2(g-1) + (g-1) = 5(g-1).$$

Their maximum bounds the spectral radius from above, and the differences

$$\begin{aligned} (g-1)^2 - (g+19) &= (g-6)(g+3), \\ (g-1)^2 - (2g+13) &= (g-6)(g+2), \\ (g-1)^2 - 5(g-1) &= (g-1)(g-6) \end{aligned}$$

are all positive for $g \geq 8$. Hence $\rho(C_g) < (g-1)^2$.

The estimate is intentionally elementary. The exact Perron root has already been identified as the dynamical degree, so the remaining task is only to place that root below a simple closed expression. The row-sum bound does this with visible factorizations and no need to solve the cubic or estimate eigenvectors. What matters is not sharpness for its own sake, but the fact that the upper bound can be checked directly from the displayed matrix entries. $\square$

$g \bmod 5$	reduced cubic	values at $t = 0, 1, 2, 3, 4$
2	$t^3 + t + 1$	1, 3, 1, 1, 4
3	$t^3 - 2t^2 - 1$	4, 3, 4, 3, 1
4	$t^3 + t^2 + t - 1$	4, 2, 3, 3, 3

Table 13: Mod-five no-root audit for the cubic Perron subfamilies. In each listed residue class the reduced cubic has no root in $\mathbb{F}_5$, hence is irreducible there.

7 Characteristic polynomial and mod-five cubic Perron subfamilies

Lemma 7.1 (Characteristic polynomial). *The characteristic polynomial of C_g is*

$$\chi_{C_g}(t) = t^3 - (2g + 11)t^2 + (g^2 - 16g + 19)t - 9(g - 1)^2.$$

Proof. For a 3×3 matrix, the characteristic polynomial is determined by the trace, the sum of principal 2×2 minors, and the determinant. Here

$$\text{tr}(C_g) = (g + 7) + 5 + (g - 1) = 2g + 11.$$

The principal 2×2 minors are

$$\det \begin{pmatrix} g + 7 & 6 \\ 2g + 4 & 5 \end{pmatrix} = 11 - 7g, \quad \det \begin{pmatrix} g + 7 & 6 \\ 2(g - 1) & g - 1 \end{pmatrix} = (g - 1)(g - 5),$$

and

$$\det \begin{pmatrix} 5 & 4 \\ 2(g - 1) & g - 1 \end{pmatrix} = -3(g - 1).$$

Their sum is

$$(11 - 7g) + (g - 1)(g - 5) - 3(g - 1) = g^2 - 16g + 19.$$

Finally,

$$\det(C_g) = \det(B_g) \det(A_g) = (-3(g - 1))^2 = 9(g - 1)^2.$$

Substituting these three quantities into the standard 3×3 characteristic-polynomial formula yields the stated expression. The coefficient pattern is rigid and uses no hidden CAS identity.

This trace–minor–determinant route is the shortest exact path from the displayed matrix to the cubic polynomial. A direct expansion of $\det(tI - C_g)$ would certainly be possible, but it would obscure the meaning of the coefficients. Here each coefficient is attached to a standard matrix invariant, so a reader can audit the cubic one invariant at a time. $\square$

Proposition 7.2 (Mod-five audit). *If $g \equiv 2, 3, 4 \pmod{5}$, then $\chi_{C_g}(t)$ is irreducible over $\mathbb{Q}$.*

Proof. Reduce the characteristic polynomial modulo 5. The three residue classes $g \equiv 2, 3, 4 \pmod{5}$ produce exactly the three cubics listed in Table 13. A cubic polynomial over $\mathbb{F}_5$ is reducible if and only if it has a root in $\mathbb{F}_5$. The table records the values at all five field elements and shows that none of the three cubics vanishes. Each reduced polynomial is therefore irreducible in $\mathbb{F}_5[t]$. Gauss’s lemma then gives irreducibility of $\chi_{C_g}(t)$ in $\mathbb{Q}[t]$.

The arithmetic input is intentionally minimal. For $g \equiv 2 \pmod{5}$ the reduced cubic is $t^3 + t + 1$; for $g \equiv 3 \pmod{5}$ it is $t^3 - 2t^2 - 1$; and for $g \equiv 4 \pmod{5}$ it is $t^3 + t^2 + t - 1$. In each case the table is already a complete no-root certificate because it records the value at every element of $\mathbb{F}_5$. No hidden finite search or external algebra system is being appealed to: the proof is exactly the five-point audit visible in the main text. $\square$

Corollary 7.3 (Cubic Perron subfamilies). *For $g \equiv 2, 3, 4 \pmod{5}$, the Perron root $\rho(C_g)$ is a degree-three algebraic integer. These are infinite cubic Perron subfamilies of the family F_g .*

Proof. Combine Proposition 7.2 with Corollary 6.3. Since $\rho(C_g)$ is the dominant eigenvalue of an integral 3×3 matrix with irreducible characteristic polynomial over $\mathbb{Q}$, it has degree three over $\mathbb{Q}$ and is a Perron algebraic integer. The word “cubic” therefore refers only to the algebraic degree of the Perron root in these residue classes; it is not a claim about a broader classification of the family. $\square$

Remark 7.4 (Why only the residues 2, 3, 4). The mod-five test is intentionally one-sided. It proves irreducibility for the listed residue classes and stays silent for $g \equiv 0, 1 \pmod{5}$. The paper does not claim that the omitted classes are reducible, nor that mod five is the only useful modulus. The corollary is exactly the arithmetic statement that the displayed table certifies and nothing broader.

8 Boundary audit, limitations, anti-claims, and conclusion

The borderline case $g = 7$ is a helpful anti-example. The seed **1** lies on the strict-cone boundary and the S -gap is positive, so the obstruction is geometric rather than a selector tie. This is why the theorem is stated for $g \geq 8$: the proof uses the open cone and the $g = 8, \dots, 11$ versus $g \geq 12$ split.

The limitations are not cosmetic. We do not claim a generic finite Newton-fan theorem, a classification of all symplectic or affine-triangular automorphisms, an entropy theorem, a periodic-point theorem, a torus theorem, a conjugacy theorem, or a priority claim. We also do not claim any CAS output as proof, nor any numerical sampling as evidence. The theorem is about one displayed family, one displayed cone, and one exact recurrence.

The bounded novelty wording should be read in the same restrained way. It is not an absolute novelty certificate. If a later comparison changed the surrounding context language, the proof of the present theorem would remain the same because it depends only on the explicit gradients, matrices, inequalities, and residue checks already displayed in the main text.

This is why the conclusion keeps returning to exactness rather than breadth. The paper succeeds by freezing one family tightly enough that every theorem-critical step fits in the main body: two gradients, six support rows, two selector gaps, one invariant cone, one visible degree row, and one finite residue audit. Any broader theorem would first need to replace at least one of those exact components with a genuinely new argument. The present manuscript does not attempt that broader project.

There is also a methodological lesson in the proof order itself. The paper never uses spectral information to discover the recurrence, never uses the final degree formula to justify visibility, and never uses the mod-five corollary to motivate the support ledger retroactively. Each stage is proved with the data available at that stage and then handed forward. In a short paper one might compress these steps into a single narrative paragraph, but for an exact theorem the separation is part of the rigor: it tells the reader which calculation depends on which earlier certification and prevents later algebra from hiding an earlier selector gap.

From the proof one should remember three facts. First, the selected matrices A_g , B_g , and C_g are read from a six-row support ledger and not from an abstract template. Second, the correct degree observable is the third q -coordinate, not a half-step intermediate. Third, the algebraic consequence of the mod-five audit is a genuine family of cubic Perron numbers, not a casual irreducibility remark.

The paper therefore closes where it began: with one exact family, one exact degree recurrence, and one exact algebraic corollary. The rest is bookkeeping around those identities. Nothing theorem-

critical is hidden offstage: a reader who wants to verify the result can replay Lemmas 3.1 through Proposition 7.2 in order and recover the whole theorem from the main body alone.

For that reason the manuscript is self-contained at the proof level: a reader only needs the fixed proof order L1 through L9, the two contextual citations, and the displayed support ledger. There is no missing appendix dependency, no figure dependency, and no computational certificate to request before the theorem can be checked line by line.

References

- Jérémy Blanc and Immanuel van Santen. Dynamical degrees of affine-triangular automorphisms of affine spaces, 2019. URL <https://arxiv.org/abs/1912.01324>.
- Enbo Shao and Xiaosong Sun. Dynamical degrees of affine-triangular automorphisms in dimension four, 2025. URL <https://arxiv.org/abs/2509.14584>.